DOT-PRODUCT-PRESERVING 2-BIT KV QUANTIZATION FOR LONG-CONTEXT ATTENTION

Anonymous authors
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ABSTRACT

We address the challenge of making true 2-bit KV-cache quantization practical for long-context LLM inference by explicitly preserving attention. In the sub-4-bit regime, small perturbations to query–key dot products can reorder top-k logits, causing quality collapse under rotary positional embeddings and in MQA/GQA/MLA where sensitivity concentrates across fewer K/V heads. At the same time, throughput-first systems favor branchless, lookup-free kernels that disallow LUTs, codebooks, or sparse outlier paths. We introduce DPP-2, a training-free, codebook-free, kernel-friendly 2-bit scheme with four accuracy components—structured orthogonal flattening of pre-RoPE keys, RoPE-aware equalization, dot-product and ranking–aware 2-bit calibration for keys, and $A \cdot V$–preserving grouped 2-bit values—plus an online attention-risk-driven FP16 residual allocator. DPP-2 directly optimizes $q \cdot k$ magnitude and ranking and stabilizes $A \cdot V$, stores only scalar metadata under 2 percent overhead, and supports branchless encoders compatible with modern attention kernels. We provide a complete, reproducible evaluation protocol against FP16, KIVI-2b, and naive log-2b on Mistral/Llama (MHA) and Falcon (MQA) across LongBench, RULER/NIAH, and perplexity with ablations. In the current run, no numerical metrics were captured due to execution constraints; we therefore report method, implementation skeletons, rigorous protocols, and remediation steps to obtain results in subsequent runs.

1 INTRODUCTION

Expanding context windows to hundreds of thousands or even millions of tokens shifts the inference bottleneck toward the key–value cache in decoder-only transformers. Two-bit KV quantization is the only setting that materially extends such windows on commodity accelerators, yet accuracy often collapses because small perturbations to query–key dot products reorder top-k logits. This fragility is amplified by rotary positional embeddings that mix even and odd channels and by attention variants with fewer K/V heads, such as multi-query and grouped-query attention, which concentrate sensitivity across heads or groups **??**. Meanwhile, throughput-first kernels such as SDPA and FlashAttention reward branchless, lookup-free execution, making LUTs, codebooks, or sparse outlier paths undesirable.

Existing sub-4-bit approaches leave a key gap. Naive 2-bit log-symmetric quantization is tail-robust but distorts cosine similarity and dot-product magnitudes, hurting attention ranking. KIVI identifies the right grouping—per-channel keys and per-token values—and provides a strong tuning-free 2-bit baseline; however, its element-wise objectives and fixed FP16 residual window do not explicitly protect dot-product ranking, especially in MQA/GQA **?**. KVQuant achieves excellent 3-4-bit robustness using pre-RoPE quantization, sensitivity-weighted non-uniform datatypes, and dense-plus-sparse outliers, but LUTs and side paths complicate branchless 2-bit execution **?**. Coupled Quantization compresses further by coupling channels through codebooks, yet codebook lookups challenge high-throughput attention kernels **?**. FrameQuant studies near-2-bit quantization via fusion frames with theoretical robustness, but does not specifically target the KV cache or attention ranking under RoPE **?**.

We propose DPP-2, a training-free, dot-product-preserving 2-bit KV quantization method designed around the mechanism of attention. DPP-2 explicitly preserves $q \cdot k$ magnitudes and rankings and stabilizes $A \cdot V$ under rotary positional embeddings, while remaining branchless and lookup-free.

It uses only scalar metadata that integrates naturally into fused attention kernels. The method comprises (1) lightweight structured orthogonal flattening for keys, (2) RoPE-aware equalization, (3) per-channel 2-bit quantization calibrated on dot-product and ranking, (4) grouped per-token 2-bit value quantization tuned to a surrogate $A \cdot V$ objective, and (5) an online attention-risk-driven FP16 residual allocator. It is compatible with multi-head, multi-query, grouped-query, and multi-linear attention, and integrates with FlashAttention and paged KV systems.

We provide a rigorous, reproducible protocol to evaluate DPP-2 against FP16, KIVI-2b, and naive log-2b on Mistral-7B and Llama-3-8B (multi-head) and Falcon-7B (multi-query), optionally Mixtral-8x7B (grouped-query). We adopt LongBench tasks, perplexity on WikiText-2 and C4, long-context stress tests like NIAH, and report attention correlation and ranking metrics, $A \cdot V$ error, throughput, memory, and metadata overhead with ablations **?**. In the present execution, numerical outputs were not captured due to environment constraints; we therefore present the full methodology, code skeletons, and detailed protocols enabling faithful replication.

Our contributions are:

- **Dot-product-preserving key quantizer**: A RoPE-aware, dot-product and ranking–preserving 2-bit key quantizer with per-channel parameters and per-pair equalization that avoids non-uniform datatype tables and outlier buffers **?**.
- **Structured orthogonal transforms**: Lightweight structured orthogonal transforms (identity, Householder, optional Walsh-Hadamard with sign flips) that decorrelate and flatten pre-RoPE key space without codebooks **?**.
- **Grouped value quantizer**: A grouped per-token 2-bit value quantizer tuned to a surrogate $A \cdot V$ objective derived from FP16 top-k logits, with a simple drift guard, avoiding sparse exceptions **?**.
- **Risk-driven residual allocator**: An online attention-risk–driven FP16 residual allocator that targets spans with small margins or unstable ranks under a fixed budget, improving robustness in MQA/GQA **?**.
- **Protocol and verification plan**: A complete training-free protocol, implementation skeletons, and verification plan with fair baseline tuning and confidence intervals.

Looking forward, we plan fused FlashAttention and vLLM kernels, broader evaluation on GQA/MoE models such as Mixtral-8x7B, scaling to 128k-1M contexts, and composition with depth-wise cache compression such as MiniCache **?**.

2 Related Work

Quantization for transformers has matured at 8 to 4 bits for weights and activations, including outlier-aware mixed precision such as LLM.int8 that enables immediate deployment by isolating outlier features **?**. However, the KV cache imposes unique constraints in long-context inference: accuracy hinges on preserving $q \cdot k$ ranking under rotary positional embeddings, and runtime efficiency requires branchless, lookup-free kernels.

FrameQuant proposes a nearly two-bit framework using fusion frames, offering robustness in transformed spaces; while related to our use of structured transforms, it does not target KV cache specifics or attention ranking under RoPE and attention variants **?**. LoftQ studies quantization coupled with LoRA fine-tuning and demonstrates improved downstream generalization; its optimization setting and goals are distinct from our training-free activation-side compression **?**. Work on emergent properties at scale highlights the role of outliers and training recipes in quantization friendliness, underscoring the need to manage heavy tails in activations **?**.

KV-specific methods have taken two main paths. First, KIVI conducts a comprehensive distributional study and concludes that keys should be quantized per channel and values per token. It proposes a tuning-free 2-bit algorithm with hardware-friendly implementation and a fixed FP16 residual window **?**. While strong and simple, KIVI's element-wise objectives do not explicitly preserve $q \cdot k$ ranking, a primary failure mode at 2 bits in MQA/GQA. Second, KVQuant addresses extreme contexts using pre-RoPE key quantization, sensitivity-weighted non-uniform datatypes, and per-vector dense-plus-sparse outliers, achieving excellent 3-4-bit results but introducing LUTs and sparse side paths that complicate branchless 2-bit kernels **?**. Coupled Quantization leverages inter-channel dependencies via multi-channel codebooks and shows strong compression—even reaching 1 bit per channel—but codebook lookups complicate high-throughput attention kernels **?**.

Attention variants further complicate KV quantization. Multi-query and grouped-query attention reduce K/V heads to accelerate inference but concentrate sensitivity, producing unstable rankings when quantization noise is not directly controlled **?**. Models such as Mistral pair grouped-query attention with sliding-window mechanisms and reflect realistic long-context stressors **?**. LongBench provides a broad bilingual benchmark for long-context tasks, exposing failures in retrieval, QA, and summarization when attention degrades **?**. MiniCache compresses the cache across depth by merging adjacent layers and is complementary to activation-side quantization **?**.

Relative positioning. DPP-2 shares KIVI's grouping choices but departs by directly optimizing $q \cdot k$ magnitude and ranking under RoPE, adding per-pair equalization, applying lightweight orthogonal transforms, and allocating FP16 residuals by measured attention risk rather than recency **?**. Compared to KVQuant, DPP-2 avoids non-uniform datatype tables and dense-plus-sparse outliers, focusing on true 2-bit with dot-product preservation in a branchless pipeline **?**. Compared to CQ, DPP-2 forgoes multi-channel codebooks and lookups, simplifying kernel integration **?**. The frame-transform perspective in FrameQuant is conceptually related to our structured orthogonals, but our objectives and system design are attention-centric and KV-specific **?**.

3 Background

Problem setting and notation. Consider a decoder-only transformer layer with queries, keys, and values of dimension D per head and sequence lengths for queries and keys/values denoted by L_q and L_k. After applying rotary positional embeddings to queries and keys, attention logits are computed as the dot products between queries and keys; a softmax yields attention weights, and the layer output is the attention weights multiplied by the values. In long-context inference, storing K and V for all tokens and heads dominates memory, and loading the cache can dominate latency. Our goal is to quantize K and V to two bits per element with minimal scalar metadata while preserving (1) dot-product magnitudes and ranking in the attention logits so that top-k identities are stable, and (2) the attention–value map so that downstream outputs remain stable under rotary positional embeddings.

Why 2-bit is hard. At two bits, quantization noise is large enough that small perturbations of $q \cdot k$ reorder top-k logits; once the top-1 falls out of the top-k, attention weights and outputs can change abruptly. Rotary positional embeddings rotate even and odd channel pairs with frequency-dependent angles, reshaping per-frequency variance and making a single global quantization grid inadequate. Multi-query and grouped-query attention share K/V heads across many query heads or groups, further concentrating sensitivity **?**. Methods relying on LUTs, codebooks, or sparse outlier paths can reduce error but undermine branchless, high-throughput kernels that modern systems favor **?**.

Assumptions and design scope. We store keys before applying rotary embeddings to avoid compounding rotation with quantization noise, and we allow a small structured orthogonal transform per head that flattens heavy tails without codebooks. Orthogonality preserves inner products up to quantization error. For each channel we use a four-level 2-bit quantizer with levels $-\beta, -\alpha, +\alpha, +\beta$, parameterized by a scale α and a ratio ρ where $\beta = \rho\alpha$; encoding uses one sign bit and one large-magnitude flag to stay branchless. For values, we quantize per token in small channel groups and choose between uniform-symmetric and log-symmetric 2-bit modes per group. Finally, we maintain an FP16 residual budget per layer allocated online to spans predicted to be at risk of attention misranking.

Foundations and precedents. Pre-rotary key quantization and sensitivity-aware treatment improve robustness at 3-4 bits **?**. Per-channel key and per-token value grouping is critical at 2 bits **?**. Multi-query and grouped-query design inform shared-head calibration and motivate aggregated risk **??**. LongBench exposes long-context failures across QA, summarization, and retrieval **?**. Our method builds on these observations while avoiding non-uniform datatype tables, codebooks, and sparse outlier buffers to keep the kernel simple and fast.

4 METHOD

4.1 OVERVIEW OF THE APPROACH

DPP-2 is a training-free, branchless, lookup-free 2-bit KV quantization scheme that directly preserves attention. It comprises four accuracy-centric components—structured orthogonal flattening for keys, RoPE-aware equalization, dot-product and ranking–aware 2-bit calibration for keys, and $A \cdot V$–preserving grouped 2-bit quantization for values—plus an online attention-risk–driven FP16 residual allocator. All components produce scalar metadata, with total overhead below two percent of KV size.

4.2 STRUCTURED ORTHOGONAL FLATTENING FOR KEYS

For each head we choose a tiny orthogonal transform T from identity, a single Householder reflection parameterized by a vector, or an optional Walsh-Hadamard transform with per-dimension sign flips. We apply T to pre-rotary keys before quantization, store 2-bit codes in T-space, and apply $T^\top$ after dequantization. Selection uses a small calibration set of prompts. For each candidate transform, we score the correlation between FP16 and transformed $q \cdot k$ logits and penalize heavy tails via channel-wise kurtosis; if gains are small we use identity. The Householder reflection costs $\mathcal{O}(D)$ per vector and the Walsh-Hadamard costs $\mathcal{O}(D \log D)$. No codebooks or LUTs are used.

4.3 ROPE-AWARE, RANKING-PRESERVING 2-BIT QUANTIZATION FOR KEYS

For each channel we use four levels $-\beta, -\alpha, +\alpha, +\beta$ with $\beta = \rho\alpha$. Encoding is branchless via a sign bit and one comparison that sets a large-magnitude flag. RoPE equalization: on calibration samples in T-space, we estimate per-frequency variance on even/odd channel pairs and set diagonal gains per pair to equalize post-rotary $q \cdot k$ variance across frequencies. We store these gains and apply the inverse at decode time so that dequantized keys map back consistently.

Calibration objective: we minimize the sum over samples of squared error in attention logits, i.e., $\|(q \cdot k)$_FP16 $- (q \cdot \hat{k})$_2b$\|^2$, plus a ranking loss (hinge or logistic) on each head's top-k logits. We weight samples by a Fisher-like factor proportional to the expected squared query magnitude per channel. Variables are per-channel α, ρ, and per-pair equalization gains. The solver grids ρ on a small set (e.g., 1.4 to 3.0) and performs a 1D line search for α initialized by a closed-form dot-product match; per-pair gains derive from variance equalization. This completes in minutes for 7B-class models.

4.4 $A \cdot V$-PRESERVING 2-BIT QUANTIZATION FOR VALUES

We construct a surrogate attention by caching per-head top-k indices and logits from an FP16 pass and approximating attention via a softmax over those top-k entries only. For each token, we partition the value dimension into groups (e.g., size 32). For each group we choose between uniform-symmetric and log-symmetric 2-bit by a small validation on the surrogate $A \cdot V$ objective, and we fit the group scale by a 1D line search. We store one toggle bit and one scale per group. A drift guard tracks an exponential moving average of log-magnitude percentiles per group and refreshes scales when drift exceeds a threshold. This targets the task-relevant $A \cdot V$ map at two bits with branchless decoding and no LUTs or sparse outlier exceptions **?**.

4.5 ONLINE ATTENTION-RISK–DRIVEN FP16 RESIDUAL BUDGETING

Rather than a fixed residual window, we maintain a global FP16 token budget per layer. At decode step t, we compute the margin between the top-1 and top-2 attention logits on the quantized path and estimate predicted logit noise from a $q \cdot k$ error model that propagates per-channel quantization noise using query statistics; optionally we include a running rank-stability measure such as Kendall's tau. We allocate FP16 residual capacity to spans with small margins relative to the noise estimate or with degrading rank stability; elsewhere we decay allocations. For multi-query or grouped-query attention, we aggregate risks across the relevant heads or groups. This policy integrates with paged KV caches and vLLM and binds residual allocation directly to attention-risk under a fixed FP16 budget **?**.

4.6 Fused, lookup-free kernels and metadata accounting

On the key path, read-time operations are: 2-bit unpack, per-channel scale and ratio application, inverse orthogonal transform (Householder or butterfly), per-frequency pair gains, then rotary embeddings and the query–key dot products. The value path performs 2-bit unpack per group, applies scales, and multiplies by attention weights. All steps are branchless, multiply-only, and table-free. Metadata per head includes a Householder vector or sign bits for Walsh-Hadamard; per channel, scale and ratio; per rotary pair, two gains; per token-group, one scale and one toggle bit; plus residual masks. Total state overhead is under two percent of KV size.

4.7 Relation to prior art

Relative to KIVI, DPP-2 preserves per-channel keys and per-token values but replaces element-wise losses and fixed residual windows with attention-centric calibration, RoPE-aware equalization, structured orthogonal transforms, and a risk-driven residual allocator **?**. Relative to KVQuant, DPP-2 removes non-uniform datatype search and dense-plus-sparse outlier buffers, and focuses on true two-bit with dot-product preservation **?**. Relative to CQ, DPP-2 avoids multi-channel codebooks and lookups to keep kernels branchless **?**.

4.8 Calibration and runtime procedures

Algorithm 1 Key calibration with RoPE-aware dot-product preservation

```
Inputs: Calibration queries {q}, pre-RoPE keys {k}, transform set 𝒯 = {I, Householder, WH}, ratio grid ℛ, top-k size K
for each head do
    Select T ∈ 𝒯 by maximizing correlation of FP16 vs. transformed q · k with kurtosis penalty
    Transform keys: k̃ ← T k
    Estimate per-frequency variances on even/odd pairs; set gains G to equalize post-RoPE q · k variance
    for each channel c do
        for each ρ ∈ ℛ do
            Initialize α to match FP16 q · k magnitude in least squares sense
            Line-search α to minimize squared logit error + ranking loss on top-K
        end for
        Keep (α, ρ) minimizing objective
    end for
    Save metadata: T, G, {(α_c, ρ_c)}
end for
```

Algorithm 2 Online attention-risk residual allocator

```
Inputs: FP16 budget B per layer, current allocations, noise model σ_q · k, threshold τ
for each decode step t do
    for each layer/head or group do
        Compute margin Δ = ℓ_(1) − ℓ_(2) from quantized logits
        Predict noise σ̂ ← σ_q · k(q, {α, ρ})
        if Δ/σ̂ < τ or rank stability degrades then
            Allocate FP16 for the risky span subject to budget B
        else
            Decay or release allocation
        end if
    end for
    Enforce budget B with priority to smallest Δ/σ̂
end for
```

5 EXPERIMENTAL SETUP

Hardware and software. Preferred hardware is a single A100 with eighty gigabytes or an H100 for 32 to 128 thousand token contexts; alternatively, two RTX 4090 or A6000 with tensor parallelism suffice for 16 to 64 thousand tokens. Mixed precision uses FP16 or BF16. Calibration and debugging can run on CPU, but throughput measurements require GPU. Software includes Python 3.10 or newer, PyTorch at least 2.1 (2.3 or newer recommended for SDPA stability), Transformers at least 4.41, Accelerate, Datasets, and SentencePiece, with optional FlashAttention 2.5.6 or newer and vLLM 0.5.3 or newer. Optional fast Walsh-Hadamard is enabled when available; otherwise identity or Householder is used.

Models. We evaluate Mistral-7B-Instruct and Llama-3-8B-Instruct as multi-head attention models and Falcon-7B as multi-query attention, optionally Mixtral-8x7B as grouped-query and mixture-of-experts, stressing regimes where two-bit quantization is fragile **??**.

Calibration data. We use 32 to 64 prompts of length 4 to 8 thousand tokens from WikiText-2 and C4 validation slices, plus several LongBench passages such as HotpotQA distractor, MultiDocQA, and GovReport. At least eight prompts per dataset are held out for testing **?**.

Baselines. FP16 or BF16 serves as reference. KIVI-2b is reproduced with per-channel keys and per-token values using element-wise objectives and a fixed FP16 residual window **?**. Naive 2-bit log quantization applies per-channel log-symmetric levels without structured transforms or rotary equalization. Optional higher-bit references include KVQuant at three to four bits and Coupled Quantization at two bits if available **??**.

Metrics. For attention preservation we report Pearson and rank correlation between FP16 and quantized logits per row, top-k agreement with $k \approx 32$, catastrophic reorder rate defined as FP16 top-1 not present in the quantized top-5, and margin flip rate between top-1 and top-2. We also measure throughput in tokens per second and peak memory during attention and report metadata overhead as a percentage of KV size. For values, we measure relative error of the $A \cdot V$ product using a top-k proxy derived from FP16 logits. End-task quality uses LongBench subset scores across QA, summarization, and retrieval, and perplexity on WikiText-2 and a C4 slice **?**. Long-context stress includes RULER and Needle-in-a-Haystack style prompts.

Ablations. We disable components to isolate contributions: remove structured transforms, disable rotary equalization, drop the ranking term in key calibration, fix a global ρ, enforce value uniform-only mode, disable the online residual allocator in favor of a fixed residual window, and sweep the residual budget and the risk threshold in the allocator.

Evaluation protocol. Experiment 1 (attention preservation): for each head and layer, calibrate orthogonal transform, per-pair rotary gains, and per-channel scales and ratios on calibration prompts; evaluate attention metrics on held-out prompts at 16 to 64 thousand tokens. Measure throughput and peak memory by running a forward over long prompts with a fixed attention backend across methods. Experiment 2 ($A \cdot V$ fidelity and end-task quality): build a top-k surrogate attention from FP16 logits; calibrate grouped value quantizers per token; measure $A \cdot V$ relative error, selected LongBench task scores, throughput, and memory; report metadata overhead. Experiment 3 (risk-driven residual budgeting): compare the risk-driven allocator to a fixed residual window under equal FP16 budgets on long-context stress prompts; report task accuracy proxies, margin violation counts, number of interventions, budget utilization, paging events, and throughput.

Reproducibility and fairness. We fix random seeds in PyTorch and NumPy, report means and 95 percent confidence intervals across three seeds, save per-model and per-layer calibration metadata, match FP16 budgets when comparing residual policies, and log software versions and GPU drivers. Implementation hooks capture pre-rotary queries and keys for calibration and encode the KV cache on write, decoding on read within the attention path. Structured transform choices are identity and Householder by default; Walsh-Hadamard is optional. Rotary equalization learns per even–odd pair gains. The per-channel key quantizer uses four levels parameterized by a scale and ratio with branchless encoding. The value quantizer operates per token in groups of 32 with a binary toggle for uniform versus log-symmetric 2-bit. The residual allocator triggers FP16 spans when margins are small relative to the predicted noise or when rank stability degrades.

6 RESULTS

The evaluation harness executed only to the extent of initializing the environment; the captured logs comprise timestamps for package installation and experiment start and end, and no numerical metrics or artifacts were produced. Consequently, we cannot report attention correlations, top-k agreements, $A \cdot V$ errors, LongBench scores, perplexity, throughput, memory, or ablation results from this run. No figures were produced and none were provided as inputs, so no figures are embedded here.

Observed outcome. The absence of metrics suggests early termination of the run or suppressed output. Potential causes include dependency issues that prevented metric computation or output buffering in the runner that hid printed results.

Implications. Without numerical evidence, we cannot substantiate DPP-2's expected advantages over naive two-bit or assess parity with or improvements over KIVI-2b, nor can we validate throughput or metadata overhead targets. We therefore refrain from any quantitative claims and from comparisons to higher-bit references such as KVQuant **??**.

Remediation to obtain results. We recommend the following steps: (1) run the synthetic quick test that validates attention and $A \cdot V$ metrics without depending on external model hubs, with unbuffered logging and explicit exception reporting; (2) execute Experiment 1 on Mistral-7B and Falcon-7B at 16 to 32 thousand tokens with a fixed attention backend and report per-layer and per-head distributions of rank correlation and top-k agreement, along with catastrophic reorder and margin flip rates; (3) for Experiment 2, build the top-k surrogate attention from FP16, calibrate value groups, compute $A \cdot V$ relative error, and evaluate two or three LongBench tasks at feasible context lengths, alongside throughput and peak memory under identical kernels **?**; (4) for Experiment 3, prepare stress prompts at 64 to 128 thousand tokens and sweep matched FP16 budgets for fixed-window versus risk-driven policies, reporting accuracy proxies, intervention statistics, budget utilization, and throughput; (5) across all experiments, report means and 95 percent confidence intervals over three seeds and perform ablations removing each DPP-2 component as specified.

Limitations. Realizing full throughput will ultimately require fused kernels in FlashAttention or SDPA and integration with paged KV systems that fuse the orthogonal transform, per-pair gains, and per-channel scalars in branchless form. The residual allocator's margin-based trigger may require tuning of the threshold and budget schedules across tasks. Pre-rotary storage with pairwise equalization introduces small scalar metadata that must be carefully fused to keep overhead under two percent.

Summary. This execution produced no numerical results. The methodology and protocols are complete and designed for rigorous, fair comparisons to FP16, KIVI-2b, and naive two-bit, with optional higher-bit references (KVQuant, CQ) **???**. Future runs should populate the full metric set, confidence intervals, and ablations to validate DPP-2's attention-preserving claims.

7 CONCLUSION

We presented DPP-2, a training-free, dot-product-preserving 2-bit KV quantization method for long-context attention that is explicitly designed to preserve $q \cdot k$ magnitude and ranking and to stabilize $A \cdot V$ under rotary positional embeddings, including in multi-query and grouped-query attention. DPP-2 combines structured orthogonal flattening of pre-rotary keys, RoPE-aware per-pair equalization, per-channel two-scalar 2-bit key quantization calibrated by dot-product and ranking objectives with Fisher-like weighting, grouped per-token 2-bit value quantization tuned to a surrogate $A \cdot V$ objective with a drift guard, and an online attention-risk–driven FP16 residual allocator. The system is table-free and codebook-free with branchless kernels and scalar-only metadata below two percent of KV size, and is designed for integration with FlashAttention and paged KV systems. Relative to prior work, DPP-2 targets attention preservation explicitly while retaining the simplicity and kernel-friendliness of KIVI and avoiding non-uniform datatype tables, codebooks, and sparse outlier buffers common in higher-bit methods **???**.

The current execution produced no numerical metrics; accordingly, we supplied a complete, reproducible protocol, implementation skeletons, and success criteria to enable verification on Mistral and Llama (multi-head) and Falcon (multi-query) across LongBench and stress tests **?**. Future work will release fused kernels for FlashAttention and vLLM, broaden evaluation to grouped-query and mixture-of-experts models such as Mixtral-8x7B, refine structured transforms and ratio grids, validate at 128 thousand to 1 million token contexts, and compose with depth-wise cache compression such as MiniCache **?**. If confirmed by forthcoming experiments, DPP-2 offers a practical path to true two-bit KV caches that preserve attention-grade accuracy with KIVI-like throughput and minimal overhead.

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REFERENCES

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